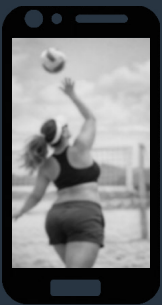
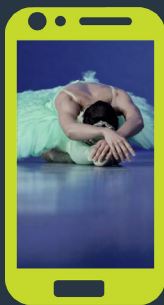
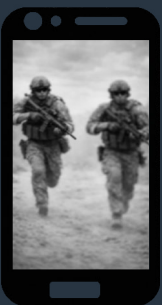
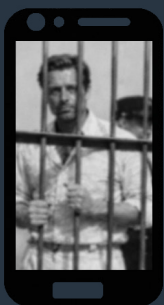
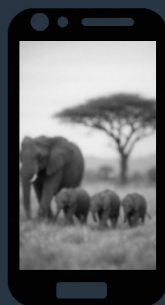
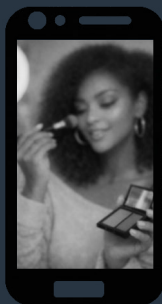
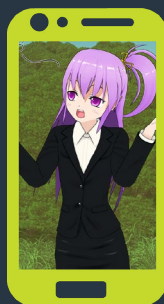
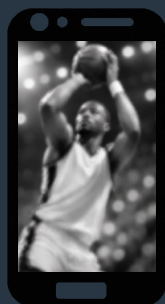


MOBILE PERSONALIZATION

The history and theory of
mobile streaming media



JUSTIN GRANDINETTI
CHARLES ECENBARGER

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Contents

<i>Introduction: Mobile Matters</i>	<i>1</i>
<i>Chapter One: The History of Mobile Personalization.</i>	<i>9</i>
<i>Chapter Two: Architectures of Mobility</i>	<i>41</i>
<i>Chapter Three: Monetizing Mobility</i>	<i>70</i>
<i>Chapter Four: The Characteristics of Mobile Streaming</i>	<i>93</i>
<i>Chapter Five: The Future of Mobile Streaming</i>	<i>121</i>
<i>Bibliography</i>	<i>128</i>

Bibliography

Abbate, Janet. *Inventing the Internet*. MIT Press, 1999.

Alphabet Inc. "Earnings 2025." Investor Relations, accessed May 12, 2025. <https://abc.xyz/investor/>.

Amin, Danial. "How Netflix Used Data to Create House of Cards: A Revolutionary Approach to Content Creation." *Medium*, March 6, 2023. <https://medium.com/@danial.a/how-netflix-used-data-to-create-house-of-cards-a-revolutionary-approach-to-content-creation-b9a114630ddc>.

Andrejevic, Mark. "The Big Data Divide." *International Journal of Communication* 8 (2014): 1673–89.

Andrejevic, Mark. *Infoglut: How Too Much Information Is Changing the Way We Think and Know*. Routledge, 2013.

Andrejevic, Mark. "Media and Mobility." In *Media History and the Foundations of Media Studies*, vol. 1 of *The International Encyclopedia of Media Studies*, edited by Angharad N. Valdivia. Blackwell, 2013.

Andrejevic, Mark. "Surveillance and Alienation in the Online Economy." *Surveillance & Society* 8, no. 3 (2010): 278–87. <https://doi.org/10.24908/ss.v8i3.4164>.

Andrejevic, Mark. "Surveillance in the Digital Enclosure." *The Communication Review* 10, no. 4 (2007): 295–317. <https://doi.org/10.1080/10714420701715365>.

Andrejevic, Mark, and Kelly Gates. "Big Data Surveillance: Introduction." *Surveillance & Society* 12, no. 2 (2014): 185–96. <https://doi.org/10.24908/ss.v12i2.5242>.

- Andrejevic, Mark, and Zala Volcic. "Seeing Like a Border: Biometrics and the Operational Image." *Digital Culture & Society* 7, no. 2 (2021): 139–58. <https://doi.org/10.25969/mediarep/21874>.
- Alper, Meryl. "Portables, Luggables, and Transportables: Historicizing the Imagined Affordances of Mobile Computing." *Mobile Media & Communication* 7, no. 3 (2018): 322–40. <https://doi.org/10.1177/2050157918813694>.
- Aoun, Nils, Chloé Currie, Ava Harrington, and Cella Wardrop. "Discover Weekly: How the Music Platform Spotify Collects and Uses Your Data." *Montreal AI Ethics Institute*, May 26, 2022. <https://montrealaiethics.ai/discover-weekly-how-the-music-platform-spotify-collects-and-uses-your-data/>.
- Auxier, Brooke, Lee Rainie, Monica Anderson, Andrew Perrin, Madhu Kumar, and Erica Turner. "Americans and Privacy: Concerned, Confused and Feeling Lack of Control over Their Personal Information." Pew Research Center, November 15, 2019.
- Awati, Rahul, and Kate Gerwig. "Net Neutrality." *TechTarget*, May 2, 2024. <https://www.techtarget.com/searchnetworking/definition/Net-neutrality>.
- Backlinko. "TikTok Statistics You Need to Know in 2025." Updated January 2026. <https://backlinko.com/tiktok-users>.
- BBC News. "Amazon and Google Deliver on TV Peace Deal." July 10, 2019. <https://www.bbc.com/news/technology-48929892>.
- Bellan, Rebecca. "The Top Social Media Apps of 2020, According to Apptopia." *Forbes*, December 3, 2020. <https://www.forbes.com/sites/rebeccabellan/2020/12/03/the-top-social-media-apps-of-2020/?sh=7beb4e5a1d4e>.
- Bennett, James. "Introduction: Television as Digital Media." In *Television as Digital Media*, edited by James Bennett and Niki Strange. Duke University Press, 2011.
- Bennett, Jane. *Vibrant Matter: A Political Ecology of Things*. Duke University Press, 2010.
- Berners-Lee, Tim, Robert Cailliau, Ari Luotonen, Henrik Frystyk Nielsen, and Arthur Secret. "The World Wide Web." *Communications of the ACM* 37, no. 8 (1994): 76–82. <https://dl.acm.org/doi/10.1145/179606.179671>.
- Blumer, Herbert, and Philip M. Hauser. *Movies, Delinquency, and Crime*. Macmillan, 1933.

- Boddy, William. "Is It TV Yet? The Dislocated Screens of Television in a Mobile Digital Culture." In *Television as Digital Media*, edited by James Bennett and Niki Strange. Duke University Press, 2011.
- Bourdon, Jérôme, and Cécile Méadel. "Inside Television Audience Measurement: Deconstructing the Ratings Machine." *Media, Culture & Society* 33, no. 5 (2011): 791–800. <https://doi.org/10.1177/01634437114047>.
- Burroughs, Benjamin. "A Cultural Lineage of Streaming." *Internet Histories* 3, no. 2 (2019): 147–61.
- Cairncross, Francis. *The Death of Distance: How the Communications Revolution Is Changing Our Lives*. Harvard Business School Press, 1997.
- Caplan, Robyn, and Tarleton Gillespie. "Tiered Governance and Demonetization: The Shifting Terms of Labor and Compensation in the Platform Economy." *Social Media + Society* 6, no. 2 (April 2020). <https://doi.org/10.1177/2056305120936636>.
- Cartier, Nicholas. "Netflix's System Architecture Represents a Rare Competitive Edge." *Streaming Media*, October 24, 2023. <https://www.streamingmedia.com/Articles/Post/Blog/Netflixs-System-Architecture-Represents-A-Rare-Competitive-Edge-161145.aspx>.
- Castells, Manuel. "Communication, Power and Counter-Power in the Network Society." *International Journal of Communication* 1 (2007): 238–66. <https://ijoc.org/index.php/ijoc/article/view/46>.
- Cedeño Montaña, Ricardo. *Portable Moving Images: A Media History of Storage Formats*. De Gruyter, 2017. <https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>.
- Childress, Sarah. "Mark Andrejevic: We Are All 'Lab Rats' Online." *Frontline*, PBS, February 18, 2014. <https://www.pbs.org/wgbh/frontline/article/mark-andrejevic-we-are-all-lab-rats-online/>.
- Cisco. "Cisco Annual Internet Report (2018–2023)." March 9, 2020. <https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>.
- Clark, Mitchell. "The Third-Party Apps Twitter Just Killed Made the Site What It Is Today." *The Verge*, January 22, 2023. <https://www.theverge.com/2023/1/22/23564460/twitter-third-party-apps-history-contributions>.

- Cohen, Mary Morley. "Always on and Always on Them': Portable Radios and the Creation of the Mobile Media Experience." *Mobile Media & Communication* 4, no. 3 (2016): 371–84. <https://doi.org/10.1177/2050157916651306>.
- Constine, Josh. "Facebook 'Watch Party' Lets Groups View Videos Simultaneously." *TechCrunch*, January 17, 2018. <https://techcrunch.com/2018/01/17/facebook-watch-party/>.
- Conti, Nicolette, Charlene Jennett, Jose Maestre, and Angela M. Sasse. "When Did My Mobile Turn into a 'Sellphone'? A Study of Consumer Responses to Tailored Smartphone Ads." In *Proceedings of the 26th Annual BCS Interaction Specialist Group Conference on People and Computers*. BCS, 2012.
- Cooney, Bill. "New Twitch Stats Reveal How Few Viewers Are Needed to Be a 'Top' Streamer." *Dexerto*, March 6, 2021. <https://www.dexerto.com/entertainment/new-twitch-stats-reveal-how-few-viewers-are-needed-to-be-a-top-streamer-1527638/>.
- Couclelis, Helen, and Harvey J. Miller. "Misses, Near-Misses and Surprises in Forecasting the Informational City." In *Societies and Cities in the Age of Instant Access*, edited by Harvey J. Miller. Springer Netherlands, 2007.
- Couldry, Nick. "Liveness, 'Reality,' and the Mediated Habitus from Television to the Mobile Phone." *The Communication Review* 7, no. 4 (2004): 353–61. <https://doi.org/10.1080/10714420490886952>.
- Covington, Paul, Jay Adams, and Emre Sargin. "Deep Neural Networks for YouTube Recommendations." In *Proceedings of the 10th ACM Conference on Recommender Systems*. Association for Computing Machinery, 2016. <https://doi.org/10.1145/2959100.2959190>.
- Cox, Christopher M. "Programming—Flow in the Convergence of Digital Media Platforms and Television." *Critical Studies in Television* 13, no. 4 (2018): 438–54. <https://doi.org/10.1177/174960201879668>.
- Crary, Jonathan. *Techniques of the Observer: On Vision and Modernity in the Nineteenth Century*. MIT Press, 2001.
- Cresswell, Tim. "Towards a Politics of Mobility." *Environment and Planning D: Society & Space* 28, no. 1 (2010): 17–31. <https://doi.org/10.1068/d1140>.
- Crofts Wiley, Stephen B., and Jeremy Packer. "Rethinking Communication After the Mobilities Turn." *The Communication Review* 13, no. 4 (2010): 263–68.

- Cubitt, Sean. "Current Screens." In *The Screen Media Reader: Culture, Theory, Practice*, edited by Stephen Monteiro. Bloomsbury, 2017.
- Davies, Hugh. "Fanology: Hand-fans in the Prehistory of Mobile Devices." *Mobile Media & Communication* 7, no. 3 (2019): 303–21. <https://doi.org/10.1177/2050157919846181>.
- de Souza e Silva, Adriana. "From Cyber to Hybrid: Mobile Technologies as Interfaces of Hybrid Spaces." *Space and Culture* 9, no. 3 (2006): 261–78. <https://doi.org/10.1177/120633120628902>.
- Deleuze, Gilles, and Felix Guattari. *A Thousand Plateaus: Capitalism and Schizophrenia*. University of Minnesota Press, 1980.
- Donner, Jonathan. *After Access: Inclusion, Development, and a More Mobile Internet*. MIT Press, 2015.
- Dreier, Troy. "Video Will Be 78% of Mobile Data Traffic by 2021, Says Cisco." *Streaming Media*, February 7, 2017. <https://www.streamingmedia.com/Articles/Articles/News/Online-Video-News/Video-Will-Be-78-of-Mobile-Data-Traffic-by-2021-Says-Cisco-116225.aspx>.
- D'Souza, Joseph. "Apple Music Statistics and Facts (2025)." *Electro IQ*, March 17, 2025. <https://electroi.com/stats/apple-music-statistics/>.
- European Union Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on Contestable and Fair Markets in the Digital Sector and Amending Directives (EU) 2019/1937 and (EU) 2020/1828 (Digital Markets Act), *Official Journal of the European Union* L265/1 (2022): 1–66. <https://eur-lex.europa.eu/eli/reg/2022/1925/oj>.
- Farman, Jason. *Mobile Interface Theory: Embodied Space and Locative Media*. Taylor and Francis, 2021.
- Federal Communications Commission. "Auction 107: 3.7 GHz Service." Accessed May 12, 2025. <https://www.fcc.gov/auction/107>.
- Fielder, Adrian. "Poaching on Public Space: Urban Autonomous Zones." In *Cinema and the City: Film and Urban Societies in a Global Context*, edited by Mark Shiel and Tony Fitzmaurice. Blackwell, 2001.
- Fortinet. "What Is Caching?" Accessed May 12, 2025. <https://www.fortinet.com/resources/cyberglossary/what-is-caching>.

- Foucault, Michel. "Questions on Geography." In *Power/Knowledge: Selected Interviews and Other Writings, 1972–1977*, edited by Colin Gordon. Pantheon, 1980.
- Frith, Jordan, and Didem Özkul. "Mobile Media Beyond Mobile Phones." *Mobile Media & Communication* 7, no. 3 (2019): 293–302. <https://doi.org/10.1177/2050157919850405>.
- Frosh, Paul. "The Face of Television." In *The Screen Media Reader: Culture, Theory, Practice*, edited by Stephen Monteiro. Bloomsbury, 2017.
- Galili, Doron. *Seeing by Electricity: Cinema, Moving Image Transmission, and the Emergence of Television, 1878–1939*. Duke University Press, 2020.
- Galloway, Alexander R. *Protocol: How Control Exists After Decentralization*. MIT Press, 2004.
- Gardner, Eriq. "Netflix Supports Net Neutrality Rules in Fight Against ISPs." *Hollywood Reporter*, April 25, 2023. <https://www.hollywoodreporter.com/business/business-news/netflix-open-internet-rules-1235792212/>.
- Gervasi, Phil. "Diving Deep into Submarine Cables: The Undersea Lifelines of Internet Connectivity." *Kentik* (blog), March 28, 2023. <https://www.kentik.com/blog/diving-deep-into-submarine-cables-undersea-lifelines-of-internet-connectivity/>.
- Geuens, Robin. "How Many Hours of Video Are Uploaded to YouTube Every Minute?" *SOAX*, October 30, 2024. <https://soax.com/research/how-many-hours-of-video-are-uploaded-to-youtube-every-minute>.
- Gillespie, Tarleton. *Custodians of the Internet: Platforms, Content Moderation, and the Hidden Decisions That Shape Social Media*. Yale University Press, 2018.
- Gillespie, Tarleton. "The Politics of 'Platforms.'" *New Media & Society* 12, no. 3 (2010): 347–64. <https://doi.org/10.1177/1461444809342738>.
- Gillespie, Tarleton. "Platforms Are Not Intermediaries." *Georgetown Law Technology Review* 2, no. 2 (2018): 198–216. <https://georgetownlawtechreview.org/wp-content/uploads/2018/07/2.2-GilesPie-pp-198-216.pdf>.
- Global Media Insight*. "YouTube Statistics 2025: Users by Country + Demographics." January 2025. <https://www.globalmediainsight.com/blog/youtube-users-statistics/>.
- Goggin, Gerard. "Mobile Web 2.0: New Imaginaries of Mobile Internet." In *Theories of the Mobile Internet: Materialities and Imaginaries*, edited by Andrew Herman, Jan Hadlaw, and Thom Swiss. Routledge, 2015.

Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2017.

Google. "Marketing Strategies." Think with Google, April–June 2018. <https://www.thinkwithgoogle.com/marketing-strategies/search/global-youtube-mobile-watch-time-statistics/>.

Google Developers. "YouTube Data API Overview." Accessed May 12, 2025. <https://developers.google.com/youtube/v3/getting-started>.

Gordon, Eric, and Adriana de Souza e Silva. *Net Locality: Why Location Matters in a Networked World*. Wiley-Blackwell, 2011.

Grandinetti, Justin, and Chris Ingraham. "Infrastructures of Flow: Streaming Media as Elemental Media." *Critical Studies in Media Communication* 39, no. 2 (2021): 92–106. <https://doi.org/10.1080/15295036.2021.2015536>.

Gu, Li, Zhenzhen Gao, and Wenjing Li. "What Drives Me to Use TikTok: A Latent Profile Analysis of Users' Motives." *Frontiers in Psychology* 13 (2022): e992824. <https://doi.org/10.3389/fpsyg.2022.992824>.

Hafner, Katie, and Matthew Lyon. *Where Wizards Stay up Late: The Origins of the Internet*. Simon & Schuster, 1996.

Hallinan, Blake, and Ted Striphas. "Recommended for You: The Netflix Prize and the Production of Algorithmic Culture." *New Media & Society* 18, no. 1 (2016): 117–37. doi:[10.1177/1461444814538646](https://doi.org/10.1177/1461444814538646).

Handley, Lucy. "Nearly Three Quarters of the World Will Use Just Their Smartphones to Access the Internet by 2025." *Yahoo! Finance*, January 24, 2019. <https://finance.yahoo.com/news/nearly-three-quarters-world-just-100800265.html>.

Haraway, Donna. *A Manifesto for Cyborgs: Science, Technology, and Socialist Feminism in the 1980s*. Center for Social Research and Education, 1985.

Haroon, Muhammad, Magdalena Wojcieszak, Anshuman Chhabra, Xin Liu, Prasant Mohapatra, and Zubair Shafiq. "Auditing YouTube's Recommendation System for Ideologically Congenial, Extreme, and Problematic Recommendations." *Proceedings of the National Academy of Sciences of the United States of America* 120, no. 50 (2023): e2213020120. <https://doi.org/10.103/pnas.2213020120>.

Hatmaker, Taylor. "Twitch Plans to Cut Subscription Revenue for Some Top Streamers in Push Toward Ads." *TechCrunch*, September 21, 2022. <https://techcrunch.com/2022/09/21/twitch-subscription-revenue-share-changes/>.

- Hay, James. "Locating the Televisual." *Television & New Media* 2, no. 3 (2001): 205–34. <https://doi.org/10.1177/152747640100200303>.
- Hect, John. "Netflix Chief Downplays Nielsen Plans to Measure Streaming Service Viewership." *Hollywood Reporter*, November 24, 2014. <https://www.hollywoodreporter.com/tv/tv-news/netflix-chief-downplays-nielsen-plans-751931/>.
- Helmond, Anne. "The Platformization of the Web: Making Web Data Platform Ready." *Social Media + Society* 1, no. 2 (2015): 1–11. <https://doi.org/10.1177/2056305115603080>.
- Hern, Alex. "Facebook Users Cannot Avoid Location-Based Ads, Investigation Finds." *Guardian*, December 19, 2018. <https://www.theguardian.com/technology/2018/dec/19/facebook-users-avoid-location-based-ads-settings-investigation-reveals>.
- History.com. "First Drive-in Movie Theater Opens." November 13, 2009. <https://www.history.com/this-day-in-history/first-drive-in-movie-theater-opens>.
- Horst, Heather A. "The Infrastructures of Mobile Media: Towards a Future Research Agenda." *Mobile Media & Communication* 1, no. 1 (2013): 147–52.
- Huhtamo, Erkki, and Jussi Parikka. "Introduction: An Archaeology of Media Archaeology." In *Media Archaeology: Approaches, Applications, and Implications*, edited by Erkki Huhtamo and Jussi Parikka. University of California Press, 2011.
- Huhtamo, Erkki. "Pockets of Plenty: An Archaeology of Mobile Media." In *The Mobile Audience: Media Art and Mobile Technologies*, edited by Martin Rieser. Rodopi, 2011.
- Ikenson, Ben. *Patents: Ingenious Inventions*. Black Dog & Leventhal, 2004.
- Ingham, Tim. "TikTok Users Watch the App for an Average of 89 Minutes per Day—and There's 732M of Them Worldwide." *Music Business Worldwide*, April 14, 2021. <https://www.musicbusinessworldwide.com/tiktok-userswatch-the-app-for-an-average-of-89-minutes-per-day-and-theres-732m-of-them-worldwide/>.
- Intel Newsroom. "Intel Study Finds 5G Will Drive \$1.3 Trillion in New Revenues in Media and Entertainment Industry by 2028." October 11, 2018. <https://newsroom.intel.com/news/intel-study-finds-5g-willdrive-1-3-trillion-new-revenues-media-entertainment-industry-2028/#gs.a3mvmj>.
- Introna, Lucas, and David Wood. "Picturing Algorithmic Surveillance: The Politics of Facial Recognition Systems." *Surveillance & Society* 2, no. 2–3 (2004). <https://doi.org/10.24908/ss.v2i2/3.3373>.

- Irani, Lilly, Robin Jeffries, and Andrea Knight. "Rhythms and Plasticity: Television Temporality at Home." *Personal and Ubiquitous Computing* 14, no. 7 (2010): 621–32. <https://doi.org/10.1007/s00779-009-0280-1>.
- Janota, Bartosz, and Robert Stephenson. "Spotify's Event Delivery: Life in the Cloud." *Engineering* (blog). Spotify R&D, November 12, 2019. <https://engineering.atspotify.com/2019/11/spotify-event-delivery-life-in-the-cloud/>.
- Jenner, Mareike. "Is This TVIV? On Netflix, TVIII and Binge-Watching." *New Media & Society* 18, no. 2 (2014): 257–73.
- Jensen, Klaus Bruhn, and Rasmus Helles. "Speaking into the System: Social Media and Many-to-One Communication." *European Journal of Communication* 32, no. 1 (2017): 16–25. <https://doi.org/10.1177/0267323116682805>.
- Kaasinen, Eija. "User Needs for Location-Aware Mobile Services." *Personal and Ubiquitous Computing* 7, no. 1 (2003): 70–79. <https://doi.org/10.1007/s00779-002-0214-7>.
- Katzenbach, Christian, and Stefan Larsson. "Imagining the Digital Society: Metaphors from the Past and Present." *Digital Society Blog*. Alexander von Humboldt Institute for Internet and Society, May 15, 2017. <https://www.hiig.de/en/imagining-the-digital-society-metaphors-from-the-past-and-present/>.
- Keim, Norman O., and David Marc. *Our Movie Houses: A History of Film and Cinematic Innovation in Central New York*. Syracuse University Press, 2008.
- Khan, Lina M. "Amazon's Antitrust Paradox." *Yale Law Journal* 126, no. 3 (2017): 710–805. <https://www.yalelawjournal.org/note/amazons-antitrust-paradox>.
- Kim, Woojin. "Television and American Consumerism." *Journal of Public Economics* 208 (2022): e104609. <https://doi.org/10.1016/j.jpubeco.2022.104609>.
- Knight, Will. "This Technique Can Make It Easier for AI to Understand Videos." *Wired*, October 15, 2019. <https://www.wired.com/story/technique-easier-ai-understand-videos/>.
- Kramer, Adam D. I., Jamie E. Guillory, and Jeffrey T. Hancock. "Experimental Evidence of Massive-Scale Emotional Contagion Through Social Networks." *Proceedings of the National Academy of Sciences of the United States of America* 111, no. 24 (2014): 8788–90. <https://doi.org/10.1073/pnas.1320040111>.
- Lamarre, Thomas. "Magic Lantern, Dark Precursor of Animation." *Animation*, 6, no. 2 (2011): 127–48. <https://doi.org/10.1177/1746847711406374>.

- Lanham, Richard A. *The Economics of Attention: Style and Substance in the Age of Information*. University of Chicago Press, 2006. <https://press.uchicago.edu/ucp/books/book/chicago/E/bo3680280.html>.
- Latour, Bruno. *Science in Action: How to Follow Scientists and Engineers Through Society*. Harvard University Press, 1987.
- Lawler, Ryan. "Netflix Will Shut Down Public API Support For Third-Party Developers On November 14." *TechCrunch*, June 13, 2014. <https://techcrunch.com/2014/06/13/netflix-api-shutdown/>.
- Lee, Hye Jin, and Mark Andrejevic. "Second-Screen Theory: From Democratic Surround to the Digital Enclosure." In *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, edited by Jennifer Holt and Kevin Sanson. Routledge, 2014.
- Lefebvre, Henri. *Everyday Life in the Modern World*. Routledge, 1984.
- Locklear, Mallory. "Facebook is Testing a Watch Party-Like Feature for Messenger." *Engadget*, November 16, 2018. <https://www.engadget.com/2018/11/16/facebook-testing-watch-party-messenger/>.
- Lotz, Amanda D. *Media Disrupted: Surviving Pirates, Cannibals, and Streaming Wars*. MIT Press, 2021.
- Manovich, Lev. "A Screen's Genealogy." In *The Screen Media Reader: Culture, Theory, Practice*, edited by Stephen Monteiro. Bloomsbury, 2017.
- Manovich, Lev. "The Poetics of Augmented Space." *Visual Communication* 5, no. 2 (2006): 219–40. <https://doi.org/10.1177/1470357206065527>.
- Marcuse, Herbert. *One-Dimensional Man: Studies in the Ideology of Advanced Industrial Society*. Beacon Press, 1964.
- Marshall, Bill. "Montreal Between Strangeness, Home, and Flow." In *Cinema and the City: Film and Urban Societies in a Global Context*, edited by Mark Shiel and Tony Fitzmaurice. Blackwell, 2001.
- Marwaha, Samir. "Sandvine's 2024 Global Internet Phenomena Report: Global Internet Usage Continues to Grow." *AppLogic Networks* (blog), April 10, 2024. <https://www.applogicnetworks.com/blog/sandvines-2024-global-internet-phenomena-report-global-internet-usage-continues-to-grow>.
- Massey, Doreen. "For Space." *Geografiska Annaler: Series B, Human Geography* 89, no. 1 (2007): 83–85.

Mayer-Schönberger, Viktor, and Kenneth Cukier. *Big Data: A Revolution That Will Transform How We Live, Work, and Think*. Houghton Mifflin Harcourt, 2013.

McCarthy, Anna. *Ambient Television: Visual Culture and Public Space*. Duke University Press, 2001.

McCarthy, Anna. "From Screen to Site: Television's Material Culture, and Its Place." In *The Screen Media Reader: Culture, Theory, Practice*, edited by Stephen Monteiro. Bloomsbury, 2017.

McLuhan, Marshall. *Understanding Media: The Extensions of Man*. MIT Press, 1964.

Medialooks. "How 5G Is Transforming Live Broadcasting and Remote Production." September 2024. <https://medialooks.com/articles/untitled-7/>.

Mehta, Ivan. "TikTok Shop Officially Launches in the US." *TechCrunch*, September 12, 2023. <https://techcrunch.com/2023/09/12/tiktok-shop-officially-launches-in-the-u-s/>.

Milne, Gemma. *Smoke & Mirrors: How Hype Obscures the Future and How to See Past It*. Robinson, 2020.

Miners, Zach. "Facebook's Location-Based Ads Pitch You on the Store You Just Walked By." *PC World*, October 7, 2014. <https://www.pcworld.com/article/2719552/facebook-locations-based-ads-can-pitch-you-on-the-store-you-just-walked-by.html>.

Monea, Alexander, and Jeremy Packer. "Media Genealogy and the Politics of Archaeology." *International Journal of Communication* 10 (2016): 3141–59.

Murphy, Kevin P. *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012.

Murray, Paul B. "Did Daylight Savings Time Make the Drive-in Theater Go Dark?" *Southern Outdoor Cinema* (blog), March 31, 2012. <http://blog.southernoutdoorcinema.com/death-of-the-drive-in-theater/>.

Musser, Charles. *Before the Nickelodeon: Edwin S. Porter and the Edison Manufacturing Company*. University of California Press, 1991.

Musser, Charles. *The Emergence of Cinema: The American Screen to 1907*. University of California Press, 1994.

Nanos, Janelle. "How Companies Use Geolocation Data to Target You." *Boston Globe*, July 2018. <https://apps.bostonglobe.com/business/graphics/2018/07/foot-traffic/>.

- Narkunas, J. Paul. "Streetwalking in the Cinema of the City: Capital Flows Through Saigon." In *Cinema and the City: Film and Urban Societies in a Global Context*, edited by Mark Shiel and Tony Fitzmaurice. Blackwell, 2001.
- NAC Editorial Team. National Association of Theatre Owners Report, 2024. <https://naconline.org/2024/08/nato-releases-exhibition-outlook-report/>.
- Netflix. "Netflix Is Now Available Around the World." *Netflix Newsroom*, January 6, 2016. <https://about.netflix.com/en/news/netflix-is-now-available-around-the-world>.
- Noble, Safiya Umoja. *Algorithms of Oppression: How Search Engines Reinforce Racism*. NYU Press, 2018.
- Novak, Matt. "A Brief History of the Videophone That Almost Was." *Tech News* (blog). Gizmodo, August 28, 2013. <https://gizmodo.com/a-brief-history-of-the-videophone-that-almost-was-1214969187>.
- Nowell-Smith, Geoffrey. "Cities: Real and Imagined." In *Cinema and the City: Film and Urban Societies in a Global Context*, edited by Mark Shiel and Tony Fitzmaurice. Blackwell, 2001.
- O'Brien, Chris. "Netflix's Reed Hastings and Ted Sarandos Explain Why Screen Size Doesn't Impact Their Shows." *Venture Beat*, May 4, 2016. <https://venturebeat.com/2016/05/04/netflixs-reed-hastings-and-ted-sarandos-explain-why-screen-size-doesnt-impact-their-shows/>.
- O'Flaherty, Kate. "All the Ways Netflix Tracks You and What You Watch." *Wired*, October 10, 2021. <https://www.wired.com/story/netflix-data-tracking-privacy/>.
- Orbach, Barak K., and Liran Einav. "Uniform Prices for Differentiated Goods: The Case of the Movie-Theater Industry." *International Review of Law and Economics* 27, no. 2 (2007): 129–53.
- O'Reilly, Tim. "What Is Web 2.0: Design Patterns and Business Models for the Next Generation of Software." *Communications and Strategies*, no. 65 (2007): 17–37. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1008839.
- Oswald, Kathleen, and Jeremy Packer. "Flow and Mobile Media: Broadcast Fixity to Digital Fluidity." In *Communication Matters: Materialist Approaches to Media, Mobility, and Networks*, edited by Jeremy Packer and Stephen B. Crofts. Wiley. Routledge, 2012.

- Packer, Jeremy, and Kathleen F. Oswald. "From Windscreen to Widescreen: Screening Technologies and Mobile Communication." *The Communication Review* 13, no. 4 (2010): 309–39.
- Packer, Jeremy. "Epistemology Not Ideology OR Why We Need New Germans." *Communication and Critical/Cultural Studies* 10, no. 2–3 (2013): 295–300. <https://doi.org/10.1080/14791420.2013.806154>.
- Parks, Lisa. "My Media Studies." *Television & New Media* 10, no. 1 (2019): 126–29.
- Parks, Lisa, Julia Velkova, and Sander de Ridder, eds. *Media Backends: Digital Infrastructures and Sociotechnical Relations*. University of Illinois Press, 2023.
- Pasquale, Frank. *The Black Box Society: The Secret Algorithms That Control Money and Information*. Harvard University Press, 2015.
- Perse, Elizabeth M. *Media Effects and Society*. Laurence Erlbaum Associates, 2001.
- Peters, John Durham. *The Marvelous Clouds: Towards a Philosophy of Elemental Media*. University of Chicago Press, 2015.
- Pietrobruno, Sheenagh. "YouTube Flow and the Transmission of Heritage: The Interplay of Users, Content, and Algorithms." *Convergence* 24, no. 6 (2018): 523–37. <https://doi.org/10.1177/1354856516680>.
- Pineda, Dhanika. "Viewers Watched 20.35 Billion Hours on Twitch in 2024: 'Stream Big' Explores Why." Central Florida Public Media, December 18, 2024. <https://www.cfpublic.org/2024-12-18/viewers-watched-20-35-billion-hours-on-twitch-in-2024-stream-big-explores-why>.
- Plantin, Jean-Christophe, Carl Lagoze, Paul N. Edwards, and Christian Sandvig. "Infrastructure Studies Meet Platform Studies in the Age of Google and Facebook." *New Media & Society* 20, no. 1 (2018): 293–310.
- Pollette, Chris. "How Streaming Video and Audio Work." *How Stuff Works*, May 1, 2024. <https://computer.howstuffworks.com/internet/basics/streaming-video-and-audio2.htm>
- Qu, Tracy. "TikTok and China Version Douyin Surpass 2 Billion Download Milestone, Underlining Continued Appeal." *South China Morning Post*, April 30, 2020. <https://www.scmp.com/tech/apps-social/article/3082285/tiktok-andchina-version-douyin-surpass-2-billion-download>.

- Qualcomm. "Everything You Need to Know About 5G." *What Is 5G?* Accessed May 12, 2025. <https://www.qualcomm.com/5g/what-is-5g>.
- Rainie, Lee. "Cable and Satellite TV Use Has Dropped Dramatically in the US Since 2015." Pew Research Center, March 17, 2021. <https://www.pewresearch.org/short-reads/2021/03/17/cable-and-satellite-tv-use-has-dropped-dramatically-in-the-u-s-since-2015/>.
- Reid, Robin T. "The History of the Drive-in Movie Theater." *Smithsonian Magazine*, May 27, 2008. <https://www.smithsonianmag.com/arts-culture/the-history-of-the-drive-in-movie-theater-51331221/>.
- Ricaurte, Paola. "Data Epistemologies, the Coloniality of Power, and Resistance." *Television & New Media* 20, no. 4 (2019): 350–65.
- Robrahn, Steve. "Drive-in Theaters Slowly Fade Out of Business." *LA Times*, July 10, 1988. http://articles.latimes.com/1988-07-10/business/fi-9345_1_drive-in-theaters.
- Roettgers, Janko. "Netflix CEO Reed Hastings: Movie Theaters Haven't Innovated Beyond Popcorn." *Variety*, March 17, 2017. <https://variety.com/2017/digital/news/netflix-ceo-reed-hastings-movie-distribution-1202010848/>.
- Russell, Alex. "YouTube Video Recommendations Lead to More Extremist Content for Right-Leaning Users, Researchers Suggest." *In Focus* (blog). UC Davis, December 13, 2023. <https://www.ucdavis.edu/curiosity/news/youtube-video-recommendations-lead-more-extremist-content-right-leaning-users-researchers>.
- Salinas, Sara. "Peak Social? The Major Social Platforms are Showing a Significant Slowdown in Users." *CNBC*, August 8, 2018. <https://www.cnn.com/2018/08/08/social-media-active-users-around-the-world.html>.
- Satin, David. "Report: 52% of U.S. Households Do Not Have Cable; Nearly 70% of Streaming Users Plan to Cycle." *The Streamable*, February 13, 2023. <https://thestreamable.com/report-52-percent-of-us-households-do-not-have-cable-nearly-70-percent-of-streaming-users-plan-to-cycle>.
- Schiffer, Michael Brian. *The Portable Radio in American Life*. University of Arizona Press, 1991.
- Seigworth, Gregory J., and Melissa Gregg. "An Inventory of Shimmers." In *The Affect Theory Reader*, edited by Melissa Gregg and Gregory J. Seigworth. Duke University Press, 2010.

- Sheller, Mimi, and John Urry. "The New Mobilities Paradigm." *Environment and Planning A: Economy and Space* 38, no. 2 (2006): 207–26. <https://doi.org/10.1068/a37268>.
- Shonk, Katie. "Streaming Toward Win-Win Negotiation: Spotify Upgrades Its Dispute Resolution with Taylor Swift." Harvard Law School, April 22, 2025. <https://www.pon.harvard.edu/daily/win-win-daily/dispute-resolution-with-spotify-taylor-swift-shakes-it-off/>.
- Slack, Jennifer Daryl, and J. MacGregor Wise. *Culture and Technology: A Primer*. 2nd ed. Peter Lang Publishing, 2015.
- Sloop, John M., and Joshua Gunn. "Status Control: An Admonition Concerning the Publicized Privacy of Social Networking." *The Communication Review* 13, no. 4 (2010): 289–308. <https://doi.org/10.1080/10714421.2010.525476>.
- Smith, Michael D., and Rahul Telang. *Streaming, Sharing, Stealing: Big Data and the Future of Entertainment*. MIT Press, 2016.
- Smythe, Dallas W. "Communications: Blindspot of Western Marxism." *Canadian Journal of Political and Social Theory* 1, no. 3 (1977): 1–27. <https://journals.uvic.ca/index.php/ctheory/article/view/13715>.
- Soja, Edward. *Postmodern Geographies: The Reassertion of Space in Critical Social Theory*. Verso, 1989.
- Sperling, Nicole. "What is Jeffrey Katzenberg's Quibi All About, and Why Should You Care?" *Vanity Fair*, June 14, 2019. <https://www.vanityfair.com/hollywood/2019/06/quibi-jeffrey-katzenberg-streaming-platform-interview>.
- Spigel, Lynn. "Introduction." In *Television After TV: Essays on a Medium in Transition*, edited by Jan Olsson and Lynn Spigel. Duke University Press, 2004.
- Spigel, Lynn. *Make Room for TV: Television and the Family Ideal in Postwar America*. University of Chicago Press, 2004.
- Spigel, Lynn. *Welcome to the Dreamhouse: Popular Media and Postwar Suburbs*. Duke University Press, 2001.
- Spotify. "The Case." Time to Play Fair (website). Accessed May 12, 2025. <https://www.timetoplayfair.com/the-case/>.
- Sriram, Akash. "Instagram to Make Up More Than Half of Meta's US Ad Revenue in 2025, Report Shows." *Reuters*, December 18, 2024. <https://www.reuters.com/>

technology/instagram-make-up-more-than-half-metas-us-ad-revenue-2025-report-shows-2024-12-18/.

Srnicek, Nick. *Platform Capitalism*. Polity Press, 2016.

Statista. "Most Popular Social Networks Worldwide as of February 2025, by Number of Monthly Active Users." February 2025. <https://www.statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/>.

Stewart, Emily. "We're Entering the Age of AI BS." *Business Insider*, March 27, 2024. <https://www.businessinsider.com/generative-ai-exaggeration-openai-nvidia-microsoft-chatgpt-jobs-investors-markets-2024-3?op=1>.

Strover, Sharon, and William Moner. "The Contours of On-Demand Viewing." In *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, edited by Jennifer Holt and Kevin Sanson. Routledge, 2014.

Terranova, Tiziana. "Cyberculture and New Media." In *The Sage Handbook of Cultural Analysis*, edited by Tony Bennett and John Frow. Sage, 2008.

Thadani, Trisha, and Rachel Lerman. "7 Times Elon Musk's Predictions About Self-Driving Cars Fell Short." *The Washington Post*, October 10, 2024. <https://www.msn.com/en-us/news/technology/7-times-elon-musk-s-predictions-about-tesla-s-self-driving-cars-fell-short/ar-AA1s1jTB>.

Thibault, Ghislain. "Streaming: A Media Hydrography of Televisual Flows." *View* 4, no. 7 (2015): 110–19.

Tolentino, Ronaldo B. "Cityscape: The Capital Infrastructuring and Technologization." In *Cinema and the City: Film and Urban Societies in a Global Context*, edited by Mark Shiel and Tony Fitzmaurice. Blackwell, 2001.

Tryon, Chuck, and Max Dawson. "Streaming U." In *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, edited by Jennifer Holt and Kevin Sanson. Routledge, 2014.

Tse, Yu-Kei. "Television's Changing Role in Social Togetherness in the Personalized Online Consumption of Foreign TV." *New Media & Society* 18, no. 8 (2016): 1547–62. <https://doi.org/10.1177/1461444814564818>.

Turner, Grame. "'Liveness' and 'Sharedness' Outside the Box." *Flow*, April 8, 2011. <http://www.flowjournal.org/2011/04/liveness-and-sharedness-outside-the-box/>.

Tussey, Ethan. "Connected Viewing on the Second Screen: The Limitations of the Living Room." In *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, edited by Jennifer Holt and Kevin Sanson. Routledge, 2014.

Twitter. "New Research: Twitter Users Love to Watch, Discover, and Engage with Video." *X Blog*, May 13, 2015. https://blog.twitter.com/en_us/a/2015/new-research-twitter-users-love-to-watch-discover-and-engage-with-video.

Uricchio, William. "Television's First Seventy-Five Years: The Interpretive Flexibility of a Medium in Transition." In *The Oxford Handbook of Film and Media Studies*, edited by Robert Kolker. Oxford Academic, 2008. Published online September 18, 2012; accessed October 16, 2024. <https://doi.org/10.1093/oxfordhb/9780195175967.013.0010>.

Uricchio, William. "The Future of a Medium Once Known as Television." In *The YouTube Reader*, edited by Pelle Snickars and Patrick Vonderau. National Library of Sweden, 2009.

van Dijck, José, Thomas Poell, and Martijn de Waal. *The Platform Society: Public Values in a Connective World*. Oxford University Press, 2018.

Verstraete, Ginette. "Media, Places, and Assemblies: From Videowalls to 'Smart' Environments." *Space and Culture* 19, no. 4 (2016): 308–18. <https://doi.org/10.1177/1206331215620996>.

Vonderau, Paul. "Beyond Piracy: Understanding Digital Markets." In *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, edited by Jennifer Holt and Kevin Sanson. Routledge, 2014.

Webb, Kevin. "Netflix Says It's More Worried About Competition from Video Games Like 'Fortnite' than Other Streaming Services." *Business Insider*, January 18, 2019. <https://www.businessinsider.com/netflix-fortnite-competition-q4-earnings-2019-1>.

Welling, David. *Cinema Houston: From Nickelodeon to Megaplex*. University of Texas Press, 2011.

Williams, Raymond. *Television: Technology and Cultural Form*. Fontana, 1974.

Williams, Robert. "Twitch Readies 2-Day Live Shopping Show for Amazon Prime Day." *Mobile Marketer*, July 1, 2019. <https://www.mobilemarketer.com/news/twitch-readies-2-day-live-shopping-show-for-amazon-prime-day/557945/>.

Wolinsky, Jacob. "Katzenberg and Whitman Discuss Their New Streaming TV Venture, NewTV." *ValueWalk*, January 31, 2020.

Zambelli, Alex. "A History of Media Streaming and the Future of Connected TV." *Guardian*, March 1, 2013. <https://www.theguardian.com/media-network/media-networkblog/2013/mar/01/history-streaming-future-connected-tv>.

Zoom. "A New Era of Growth: How the Zoom Partner Ecosystem Is Evolving." *Zoom Blog*, October 2024. <https://www.zoom.com/en/blog/a-new-era-of-growth-how-the-zoom-partner-ecosystem-is-evolving/>.

Zuboff, Shoshana. *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. PublicAffairs, 2019.